

*Report of the Project Done in Partial Fulfilment of the Requirements for the Award of
Degree of Bachelor of Technology in Electronics and Communication Engineering*

WIRELESS ELECTRODE FOR BIO-SIGNAL ACQUISITION

PROJECT PHASE II REPORT

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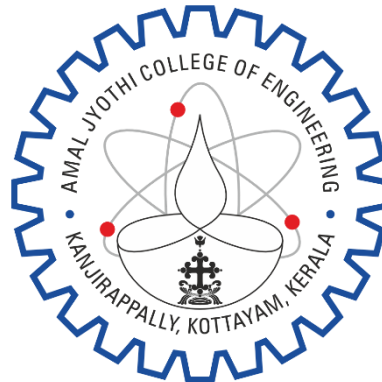
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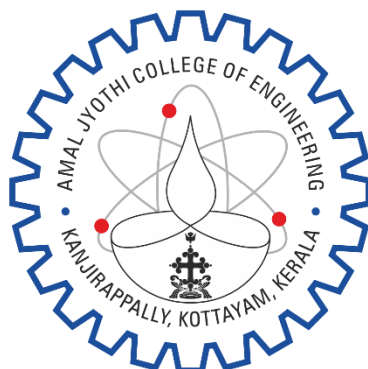
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BONAFIDE CERTIFICATE

This is to certify that the project phase II report entitled “**Wireless Electrode for Bio-signal Acquisition**” is a Bonafide report of the Eighth Semester Project Phase II [ECD416] done by **AFILLAL FRANCIS Reg No: AJC21EC003, AMEER MUHAMMED P S Reg No: AJC21EC019, DHANISH MAHMOOD K Reg No: AJC21EC040, JEEVAN JOHN Reg No: AJC21EC045** in Partial fulfilment of requirements for the award of degree of Bachelor of Technology in Electronics and Communication Engineering from APJ Abdul Kalam Technological University, on April 2025. They have done the phase II of the project with prior approval from the department.

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ABSTRACT

An advanced wireless, low-power EEG acquisition system has been developed to overcome the limitations of traditional wired configurations while ensuring exceptional signal fidelity. Prior to this development, an initial wired EEG system was constructed for parameter testing and analysis, which provided valuable insights that informed the current design. The design utilizes AD623 instrumentation amplifiers configured to provide an approximate gain of 100 and a minimum common-mode rejection ratio (CMRR) of 90 dB, thereby amplifying differential EEG signals captured from two active electrodes and a reference electrode. AC coupling at the electrode inputs is implemented using 330 nF capacitors, which, in combination with the inherent 1 M Ω electrode–skin impedance, forms a high-pass filter with a cutoff frequency near 0.48 Hz. This is complemented by robust ESD protection and comprehensive shielding techniques that further secure signal integrity.

Subsequent signal conditioning is achieved through an AD8603-based active filtering stage that incorporates a Sallen–Key low-pass filter followed by a non-inverting amplifier, collectively providing an additional nominal gain of approximately 40x and resulting in an overall system gain of roughly 4000. Digital gain adjustment is facilitated by an AD5165 digital potentiometer, enabling flexible tuning via SPI communication from a Seeed XIAO nRF52840 microcontroller. Additionally, a Driven Right Leg (DRL) circuit actively cancels common-mode interference by feeding an inverted common-mode voltage back into the body, effectively mitigating external noise sources such as mains interference.

The conditioned EEG signal is then digitized using a 10-bit ADC integrated within the Seeed XIAO nRF52840, which also incorporates a BLE module for real-time wireless data transmission. Operating at 3.3 V and drawing approximately 12 mA, the system is capable of continuous operation for over 10 hours on a modest battery, making it highly suitable for portable neurotechnology applications. Validation experiments, including EEG recordings conducted both before and after Cranial Electrical Stimulation (CES), confirm the system's capacity for high-fidelity, real-time monitoring. A Python-based graphical user interface (GUI) further facilitates effective data visualization and analysis, underscoring the system's potential in neuroscience research, clinical diagnostics, brain-computer interfaces, and neurofeedback applications.

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CHAPTER 1

INTRODUCTION

Electroencephalography (EEG) is a critical non-invasive method for recording the brain's electrical activity, capturing microvolt-level signals produced by neural events. These signals, detected via electrodes on the scalp, are inherently weak and require precise signal conditioning to preserve their fidelity. Traditional EEG systems, which rely on wired electrode configurations, often face issues such as signal degradation over long cables, motion-induced artifacts, and discomfort from gel-based electrodes. These limitations hinder extended monitoring and reduce data quality, prompting a shift toward wireless, low-power acquisition platforms that integrate advanced circuitry for improved performance.

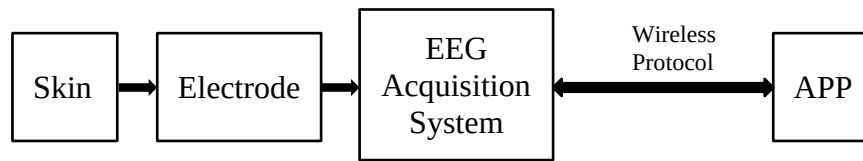


Figure 1. Wireless bio-signal acquisition model

In a modern EEG acquisition system, the signal chain begins with the electrode interface. Electrodes are designed to detect subtle neural signals while contending with the inherent electrode–skin impedance, which can compromise signal integrity if not managed properly. The initial stage employs a high-pass filter (HPF) to remove unwanted direct current (DC) offsets and low-frequency drifts. This is typically achieved using a coupling capacitor in combination with the electrode–skin impedance, resulting in an RC filter that effectively preserves the frequency components relevant to neural activity (Figure 1) [1].

After high-pass filtering, the extremely low-amplitude EEG signal is boosted by a pre-amplifier. This stage is essential for elevating the microvolt-level signals to levels suitable for further processing while minimizing the introduction of noise. The pre-amplified signal is then processed through a low-pass filter (LPF) to attenuate high-frequency noise, such as electromagnetic interference. The LPF is carefully tuned to the neural signal band, ensuring that only frequencies associated with brain activity are retained. This dual-filtering approach—removing both low-frequency drift and high-

frequency noise—ensures that the signal remains both clean and representative of the underlying neural processes [6].

Following the filtering stages, a secondary amplification stage is implemented to further enhance the signal amplitude. This stage is calibrated to provide additional gain, ensuring that the conditioned signal fully utilizes the dynamic range of the analog-to-digital converter (ADC) without causing saturation or distortion. The ADC, typically with a resolution of 12 bits or higher, digitizes the analog signal so that it can be processed by digital systems. The conversion process is critical, as it captures the fine variations in the EEG signal and prepares the data for storage, analysis, or real-time processing.

The integration of wireless, low-power technology in EEG systems has significantly advanced the field. Modern designs incorporate compact, battery-operated modules that enable continuous monitoring and real-time data transmission without the constraints imposed by wired systems. These wireless modules combine the entire signal processing chain from electrode capture through filtering, amplification, and digitization into a single, efficient unit. Advanced techniques such as active noise cancellation, where an inverted noise signal is fed back into the system to cancel interference, are employed to further enhance signal quality. This comprehensive integration ensures that even in environments with substantial electromagnetic interference, the EEG signals are recorded with high precision and reliability.

The engineering challenges in developing a robust EEG acquisition circuit lie in balancing sensitivity and noise reduction. Each stage of the signal chain must be optimized to maintain the integrity of the neural signals. The electrode stage must provide reliable contact with the scalp while minimizing impedance-induced signal loss. The filtering stages are designed to precisely remove unwanted frequency components without attenuating the neural signals. The amplification stages must boost the signal sufficiently while avoiding the introduction of additional noise or distortion. Finally, the ADC must capture the conditioned signal with high resolution, enabling accurate digital representation and subsequent analysis.

Wireless EEG systems not only improve patient comfort and mobility but also enhance data reliability by eliminating issues associated with long cable runs and motion artifacts. The ability to monitor brain activity continuously over extended periods has significant clinical implications, particularly for patients with conditions such as epilepsy or sleep disorders who require long-term monitoring. Additionally, these

systems facilitate research in real-world environments, where controlled laboratory conditions are not feasible. The portability and ease-of-use of modern EEG acquisition devices make them ideally suited for applications in brain–computer interfaces and neurofeedback, where real-time data analysis and immediate feedback are essential.

Advancements in microelectronics and materials science continue to drive improvements in EEG technology. Future developments are expected to yield electrodes with lower impedance and better biocompatibility, further enhancing signal quality. Similarly, improvements in low-noise amplifiers and high-resolution ADCs will enable more precise capture of neural signals. The integration of machine learning algorithms for real-time artifact rejection and data analysis holds promise for automating and refining the interpretation of EEG data. These innovations will not only narrow the gap between laboratory-grade systems and portable devices but also expand the range of applications in both clinical and consumer domains.

In summary, the evolution from traditional wired EEG systems to modern, wireless, low-power acquisition platforms represents a significant leap in neurotechnology. By meticulously designing each stage from electrode capture and high-pass filtering through pre-amplification, low-pass filtering, secondary amplification, and analog-to-digital conversion—engineers have developed systems that capture neural signals with exceptional fidelity. The transition to wireless technology has further improved system usability, enabling continuous, real-time monitoring in diverse environments. These advancements enhance the capabilities of EEG systems, facilitating more accurate clinical diagnostics, robust research applications, and innovative developments in brain–computer interfacing and neurofeedback. As research and technology progress, these systems will continue to play a pivotal role in advancing our understanding of brain function and improving patient care.

CHAPTER 2

MOTIVATION AND PROBLEM STATEMENT

2.1. MOTIVE OF THE PROJECT

The motive of this project is to develop a next-generation wireless EEG electrode system that overcomes the significant limitations of traditional EEG setups by leveraging modern digital and sensor technologies. Conventional systems are often burdened by bulky, wired configurations that restrict patient movement and lead to signal degradation from long cables and motion-induced artifacts. In contrast, this project aims to deliver a highly portable and user-friendly solution capable of high-fidelity, real-time monitoring of brain activity in dynamic and ambulatory environments. By integrating Bluetooth Low Energy (BLE) for efficient wireless data transmission, the system eliminates cumbersome cables and supports continuous monitoring in clinical, home, or outdoor settings.



Figure 2. Traditional EEG monitoring system

Advanced signal preprocessing techniques are employed to amplify the weak EEG signals while improving the signal-to-noise ratio (SNR) and enhancing the common-mode rejection ratio (CMRR), ensuring that the subtle electrical activity of the brain is captured accurately even in the presence of environmental noise. Ultimately, the project is driven by the need for an EEG system that is not only technically robust achieving high gain, SNR, and CMRR but also versatile enough for a broad range of applications, including clinical diagnostics, neurofeedback, brain-computer interfaces (BCIs), and cognitive state monitoring. This integrated approach promises to expand the utility of

EEG by making it more accessible, accurate, and practical for real-world applications, thereby advancing both healthcare and neuroscience research.

2.2. OBJECTIVE

Objectives of the Study

The objective of this project is to design and implement a compact, wireless, and low-power EEG acquisition system that effectively addresses the limitations of conventional wired EEG devices. The system integrates a precision instrumentation amplifier for initial differential signal capture and an active filtering stage, which includes a low-pass filter paired with a digitally adjustable non-inverting amplifier. Achieving an overall gain of approximately 4000 is critical, as it amplifies microvolt-level neural signals into a range that is readily processed and analyzed. This high gain is essential for preserving the fine details of brain activity, while the inclusion of a digital potentiometer allows for precise fine-tuning of the gain to adapt to varying signal conditions.

Furthermore, the system employs advanced design strategies such as robust ESD protection, efficient AC coupling, and optimal biasing techniques to minimize interference from motion artifacts, muscle noise, and external electrical disturbances. Bluetooth Low Energy (BLE) is utilized to enable low-power, real-time wireless data transmission, eliminating the need for cumbersome wiring and significantly enhancing user mobility and comfort [12]. A dedicated application has been developed to connect with the EEG device, offering seamless real-time visualization and analysis of the acquired signals. In summary, this project aims to deliver a high-fidelity, portable, and user-friendly EEG solution that is suitable for long-term monitoring, neurofeedback, brain-computer interface (BCI) applications, and other real-world ambulatory uses, thereby advancing EEG technology in both clinical and research settings.

CHAPTER 3

LITERATURE SURVEY

- [1]. Zhiyuan Yu, Shengwen Guo, “*A low-cost, wireless, 4-channel EEG measurement system used in virtual reality environments*”, *HardwareX*, Elsevier, 2024.

This study presents NeuroVista, a low-cost, compact, and wireless 4-channel EEG measurement system designed for frontal area monitoring. Priced at 60.07 USD, it is compatible with VR headsets, enabling EEG measurements in VR environments. The system offers high precision with features such as low noise, a high common mode rejection ratio, and a resolution under $0.1\mu\text{V}$. It includes metal dry electrodes, filters to reduce noise, and can be customized for different research needs. NeuroVista provides an affordable, user-friendly solution for EEG measurements in neuroscience and psychology research. <Body text; Font: Times New Roman 12 Normal Justified>

- [2]. Khizar Anjum, Dario Pompili, “*Battery-Less Implantable Continuous EEG Monitoring via Anisotropic Diffusion*”, *IEEE Journal on Selected Areas in Communications*, Vol. 42, No. 8, 2024.

This article introduces an innovative approach for ultra-low-power hybrid analog-digital processing of multimodal physiological data, with a focus on EEG signals. The authors propose a novel analog Convolutional Processing Unit (CvPU) that uses anisotropic diffusion in electrical circuits for convolution, setting it apart from traditional methods. A controller architecture is also presented, enabling the sequential execution of multiple convolutional layers using the same CvPU array. The system is applied to seizure detection using EEG signals, evaluated on a clinical dataset. The performance and feasibility of the CvPU-based convolution were assessed through SPICE simulations. The study also explores the scalability of the approach for battery-less and implantable applications, comparing it with digital and hybrid analog-digital methods in terms of power and space efficiency.

- [3].Ijaz Ahmad, Mingxing Zhu, Guanglin Li, Danish Javeed, Prabhat Kumar, Shixiong Chen, “A Secure and Interpretable AI for Smart Healthcare System: A Case Study on Epilepsy Diagnosis Using EEG Signals”, IEEE Journal of Biomedical and Health Informatics, Vol. 28, No. 6, June 2024.

This study introduces an XAI-based system for automated epileptic seizure detection (ESD) in EEG signals, incorporating feature extraction, seizure detection, and explainable decision-making. It utilizes the Butterworth filter for artifact removal, DTCWT for signal decomposition, and selects features using correlation methods. The features are processed with Stacking Ensemble Classifiers (SEC) for seizure detection, and Shapley Additive Explanations (SHAP) provide interpretability. Blockchain ensures data security. The system shows a 2% improvement in accuracy, recall, specificity, and F1-score across multiple EEG datasets, enhancing diagnostic accuracy while ensuring data privacy in smart healthcare.

- [4].Kunpeng Gao,Le Li, Bowen Ji, Nailong Wu, “Pin-shaped Ag/AgCl fabric electrode coated with hydrogel for EEG recording in hairy areas”, IEEE Sensors Journal, 2024.

This paper introduces a pin-shaped fabric electrode designed for EEG recording, especially in areas with hair, achieving low contact impedance. The electrode, made from Ag/AgCl-coated fabric and polyurethane columns, is filled with hydrogel to further reduce impedance. It combines the advantages of dry pin-shaped and fabric electrodes, with a contact impedance of $10.9 \pm 2.7 \text{ k}\Omega$, lower than most hydrogel or semi-dry electrodes. The hydrogel is protected within the fabric, ensuring durability, and the electrode's performance remains stable over time, with an average impedance under $15 \text{ k}\Omega$ after 30 days. Its EEG signal quality is comparable to wet electrodes, offering a practical and long-lasting solution for EEG monitoring.

- [5].Hafza Ayesha Siddiq, Zhenning Tang, Yan Xu, Laishuan Wang, Muhammad Irfan, Saadullah Farooq Abbasi, Anum Nawaz, Chen Chen, Wei Chen, “Single-Channel EEG Data Analysis Using a Multi-Branch CNN for Neonatal Sleep Staging”, IEEE Access, February 2024.

This study investigates optimal electrode configurations for neonatal sleep staging, aiming to reduce technical complexities and skin irritation risks. Using single-channel EEG data and a Multi-Branch Convolutional Neural Network (CNN), the model classified neonatal sleep states from 16,803 30-second segments. A combination of 74 time and frequency features improved performance, with principal component analysis (PCA) identifying key features. The F3 channel provided the best results, with 74.27% accuracy and a kappa of 0.61. Four left-side electrodes slightly outperformed right-side electrodes. The research highlights the role of specific EEG channels in sleep classification and shows that using single-channel EEG can reduce computational load while maintaining strong accuracy, aiding early detection of sleep-related issues in neonates.

[6].Alireza Dabbaghian, Yoland El-Hajj, Milad Ghalamboran, Gerd Grau, Hossein Kassiri, “A 0.67 μ V-IIRN super- $T\Omega$ -ZIN 17.5 μ W/Ch Active Electrode With In-Channel Boosted CMRR for Distributed EEG Monitoring”, IEEE Transactions on Biomedical Circuits and Systems, Vol. 18, No. 1, February 2024.

This paper details the design and development of an active electrode (AE) integrated circuit (IC) for wearable EEG recording. The AE features double common-mode rejection (CMRR), which ensures stable signal quality regardless of electrode-to-electrode variations. Its DC-coupled architecture provides extremely high input impedance, making it less sensitive to impedance changes at the electrode-skin interface—a crucial advantage for dry-electrode systems. The AE also performs signal quantization and serialization on-chip, improving scalability by using a single data bus for communication across all electrodes. Additional features include continuous monitoring of electrode-skin connection stability, dynamic adjustment of input DC levels, and a feedback loop to further enhance CMRR. The paper also discusses the development of printed tattoo electrodes that offer excellent bio-compatibility, stability, and flexibility, and are compatible with the AE system. The AE IC, designed using 180 nm CMOS technology, demonstrates an impressive CMRR of 82.2 dB, a voltage gain of 52.8 dB, and operates at a very low power consumption of 17.5 μ W per channel. The system has been validated for both EEG and ECG recording through in-vivo tests.

- [7]. Wenjie Cui, Kai Lin, Guangda Liu, Yubing Sun, Jing Cai, “*A Wireless Integrated EEG-fNIRS System for Brain Function Monitoring*”, *IEEE Sensors Journal*, Vol. 24, No. 2, 15 January 2024.

The proposed integrated EEG-fNIRS system enables simultaneous, real-time monitoring of brain activity and hemodynamic changes in mobile conditions. Its scalable sensor cap allows flexible configurations for various experiments. The system meets noise and drift requirements, with results consistent with existing literature in visual evoked experiments. While promising for research and clinical use, further testing is needed to ensure reliability across different applications.

- [8].Fuze Tian, Lixian Zhu, Qiuxia Shi, Rui Wang, Lixin Zhang, Qunxi Dong, Kun Qian, Qinglin Zhao, Bin Hu, “*The Three-Lead EEG Sensor: Introducing an EEG-Assisted Depression Diagnosis System Based on Ant Lion Optimization*”, *IEEE Transactions on Biomedical Circuits and Systems*, Vol. 17, No. 6, December 2023.

This study presents a wearable three-lead EEG sensor for depression diagnosis, offering a portable and non-invasive alternative to traditional methods. The sensor, with flexible electrodes for prefrontal EEG data collection, demonstrated low noise, high signal-to-noise ratio, and minimal impedance. Using data from 70 depressed patients and 108 healthy controls, feature selection was optimized through the Ant Lion Optimization algorithm, and the k-NN classifier achieved a high accuracy of 90.7%. However, the reliance on a single classifier and the limited three-lead configuration may restrict its effectiveness compared to more comprehensive setups. While promising, further research is needed to test the system's scalability, robustness, and real-world applicability for widespread use in clinical settings.

- [9].Alireza Dabbaghian, Hossein Kassiri, “*An 8-Channel Ambulatory EEG Recording IC With In-Channel Fully-Analog Real-Time Motion Artifact Extraction and Removal*”, *IEEE Transactions on Biomedical Circuits and Systems*, Vol. 17, No. 5, October 2023.

This paper presents an 8-channel EEG recording IC with an innovative on-chip motion artifact extraction and removal method, implemented in the analog domain using a dual-path feed-forward approach. Unlike traditional systems that rely on DSP units and high-DR ADCs, this design enables stand-alone channel operation with minimal power

consumption (55 μ W per channel). The system achieves significant motion artifact suppression (41.5 dB) while maintaining a 48.3 dB amplification gain and a 300 Hz bandwidth. Although the system shows promising results in artifact suppression, its scalability and low power consumption make it an efficient solution for wearable EEG applications in dynamic environments.

- [10]. Antonello Mascia, Riccardo Collu, Andrea Spanu, Matteo Frascini, Massimo Barbaro, Piero Cosseddu, “Wearable System Based on Ultra-Thin Parylene C Tattoo Electrodes for EEG Recording”, MDPI, 2023.**

This study introduces a wearable EEG system using a custom low-power PCB and ultra-conformable Parylene-C tattoo electrodes. It was tested in a rest-state experiment and compared to the commercial Muse headset, showing similar performance. The results highlight the potential of using tattoo electrodes in portable systems for unobtrusive brain activity monitoring, offering a promising approach for applications in sports, fitness, and biomedical fields. However, further validation in diverse settings is needed to fully assess its practical use and reliability.

- [11]. Filippo Battaglia, Associate Member, IEEE; Giovanni Gugliandolo, Member, IEEE; Giuseppe Campobello, Member, IEEE; and Nicola Donato, Senior Member, IEEE, “EEG-Over-BLE: A Low-Latency, Reliable, and Low-Power Architecture for Multichannel EEG Monitoring Systems”, IEEE, August 2023.**

This article introduces a novel EEG-over-Bluetooth Low Energy (BLE) architecture specifically designed for multichannel EEG monitoring while emphasizing low power consumption. The system incorporates a low-power analog front-end (AFE) and a Bluetooth 5.0-compatible system-on-chip (SoC), along with a near-lossless EEG compression algorithm and a straightforward channel encoding scheme to enhance energy efficiency and network performance. Evaluations demonstrate that the system, powered by commercial small-form-factor batteries, supports a 64-channel configuration for over 86 days, achieving a packet loss rate below 0.1% and latency under 35 ms, in compliance with ISO/IEEE 11073 standards for medical devices. This architecture is particularly well-suited for long-term, lightweight, and efficient health monitoring applications.

- [12]. Congying He, Yu-Yi Chen; Chun-Ren Phang, Cory Stevenson, I-Ping Chen; Tzzy-Ping Jung, and Li-Wei Ko, “*Diversity and Suitability of the State-of-the-Art Wearable and Wireless EEG Systems Review*”, Aug 2023.

This review examines the evolution and applications of wireless EEG systems, emphasizing their growing accessibility and importance in research, consumer, and clinical settings. It presents a comparative analysis of 16 commercially available systems by evaluating key parameters such as the number of channels, sampling rate, cost, battery life, and resolution. The findings indicate that affordability and ease of use are crucial for consumer applications, while FDA/CE-certified systems are best suited for clinical environments, and high-density raw EEG data systems are ideal for research purposes. The article establishes selection criteria based on specific application needs and provides an overview of current system specifications, highlighting their potential to drive innovation and advance wireless EEG research and applications.

- [13]. Zhuangzhuang Wang, Nan Zhao, Gencai Shen, Chunpeng Jiang, Jingquan Liu, “*MEMS-Based Flexible Wearable Tri-Polar Concentric Ring Electrode Array With Self-Adhesive Graphene Gel for EEG Monitoring*”, IEEE Sensors Journal, Vol. 23, No. 3, 1 February 2023.

This study focuses on enhancing EEG signal analysis by addressing the spatial resolution limitations of traditional electrodes. While EEG provides excellent temporal resolution, its spatial resolution is hindered by the blurring effects of the head volume conductor. To overcome this, the researchers designed a tri-polar concentric ring electrode (TCRE) array, which offers better spatial resolution than conventional electrodes. To further improve electrode performance, a flexible, self-adhesive graphene gel was introduced to reduce contact impedance, achieving values below 10 k Ω at 10 Hz. The system successfully recorded alpha oscillations, with the TCRE showing a high coherence of 0.913 compared to commercial Ag/AgCl electrodes. Additionally, the TCRE demonstrated significantly lower mutual information between positions, indicating its superior spatial resolution over traditional disk electrodes, all while being comfortable to wear due to its flexible material.

- [14]. **Filippo Battaglia, Giovanni Gugliandolo, Giuseppe Campobello, Nicola Donato,** “*EEG-Over-BLE: A Low-Latency, Reliable, and Low-Power Architecture for Multichannel EEG Monitoring Systems*”, **IEEE Transactions on Instrumentation and Measurement, Vol. 72, 2023.**

This article introduces a low-power, multichannel EEG-over-Bluetooth Low Energy (BLE) system designed for long-term health monitoring. The system incorporates a low-power analog front-end (AFE) for EEG signals and a system-on-chip (SoC) that supports Bluetooth 5.0. To optimize energy efficiency and network performance, the design includes a near-lossless EEG compression algorithm and channel encoding. Simulation results show that the system can last over 86 days with commercial small-form-factor batteries for a 64-channel EEG setup. Additionally, the system is highly reliable, with a packet loss rate below 0.1% and latency under 35 ms, meeting medical communication standards.

- [15]. **Wei-Yen Hsu, Ya-Wen Cheng,** “*EEG-Channel-Temporal-Spectral-Attention Correlation for Motor Imagery EEG Classification*”, **IEEE Transactions on Neural Systems and Rehabilitation Engineering, Vol. 31, 2023.**

The WTS-CC method proposed in this study enhances motor-imagery EEG signal discrimination in brain-computer interfaces by addressing limitations in existing models that neglect key spatial, temporal, and spectral features. The approach combines three modules: temporal feature extraction (iTFE), EEG-channel attention (DEC), and wavelet-based temporal-spectral attention (WTS) to extract more discriminative features. Experimental results show that WTS-CC outperforms current methods in classification metrics, including accuracy and AUC, across multiple datasets. However, the method's complexity could impact real-time performance, and further testing is needed to assess its effectiveness in diverse, practical BCI applications.

- [16]. **G. L. Cerone, A. Giangrande, M. Ghislieri, M. Gazzoni, H. Piitulainen, A. Botter,** “*Design and Validation of a Wireless Body Sensor Network for Integrated EEG and HD-sEMG Acquisitions*”, **IEEE Transactions on Neural Systems and Rehabilitation Engineering, Vol. 30, 2022.**

This article introduces a fully wireless body sensor network for acquiring both EEG and high-density surface EMG (HD-sEMG) signals, designed to study sensorimotor

integration. The system includes wireless bio-signal acquisition and synchronization modules for time-locked recordings. It was characterized for synchronization accuracy and signal quality, with a comparison to a wired EEG device showing its effectiveness. This wireless system enables integrated signal acquisition for dynamic studies of sensorimotor control, overcoming the limitations of traditional wired technologies.

- [17]. Gencai Shen, Kunpeng Gao, Nan Zhao, Zhuangzhuang Wang, Chunpeng Jiang, Jingquan Liu, ***“A Fully Flexible Hydrogel Electrode for Daily EEG Monitoring”***, **IEEE Sensors Journal**, Vol. 22, No. 13, July 1, 2022.

This study introduces a flexible hydrogel electrode for daily EEG monitoring, addressing the complexity and discomfort of traditional wet electrodes. The hydrogel electrode, enhanced with glycerol for moisture retention, maintains elasticity over time and provides easy installation. It effectively captures EEG signals, showing high coherence (0.910) with conventional Ag/AgCl wet electrodes. While promising for daily use, the electrode's long-term performance and scalability in diverse applications remain areas for further evaluation.

- [18]. Hongfei Wang, Jingjing Wang, Da Chen, Song Ge, Yijian Liu, Zhengjie, Zhang Wang, Xiaojun Zhang, Qiuquan Guo, Jun Yang, ***“Robust Tattoo Electrode Prepared by Paper-Assisted Water Transfer Printing for Wearable Health Monitoring”***, **IEEE Sensors Journal**, Vol. 22, No. 5, March 1, 2022.

The study presents a paper-assisted strategy for creating tattoo ECG electrodes using water transfer printing, offering a low-cost, flexible, and robust solution for long-term bioelectrical signal monitoring. These tattoo electrodes conform to the skin, accommodate deformations, and provide high-quality ECG signals with strong resistance to motion artifacts, even during activities like exercise. The system integrates with a wireless setup, enabling continuous ECG monitoring via Bluetooth. While promising, further testing on long-term durability, power consumption, and broader applicability across diverse user groups is needed to validate its full potential in real-world use.

- [19]. Wei-En Lee, Chieh Tsou, Zhan-Xian Liao, Po-Yen Lu, Tsung-Hsien Lin, Shuenn-Yuh Lee, Chou-Ching K. Lin, and Gia-Shing Shieh, ***“Intelligent Wireless EEG Measurement System With Electrical and Optogenetic Stimulation for Epileptic***

Seizure Detection and Suppression”, IEEE Engineering in Medicine and Biology Society Section, June 8, 2021.

This study introduces a compact, two-channel EEG system that integrates both electrical and optical stimulation for real-time seizure detection and suppression. Managed wirelessly, the system utilizes a Matlab-based interface to monitor intracranial signals and adjust stimulation parameters such as intensity, frequency, duty cycle, and duration. Validated on a small printed circuit board and tested in drug-induced seizure models in mice, the system effectively suppressed epileptic events, indicating its potential as a promising approach for advanced, closed-loop epilepsy management.

- [20]. Wei-En Lee, Chieh Tsou, Zhan-Xian Liao, Po-Yen Lu, Tsung-Hsien Lin, Shuenn-Yuh Lee, Chou-Ching K. Lin, Gia-Shing Shieh, “*Intelligent Wireless EEG Measurement System with Electrical and Optogenetic Stimulation for Epileptic Seizure Detection and Suppression*”, IEEE Access, IEEE Engineering in Medicine and Biology Society Section, May 2021.

This paper introduces a compact two-channel EEG measurement system designed for seizure detection and suppression using electrical and optical stimulations. Controlled through a wireless interface and Matlab GUI, the system provides real-time monitoring of intracranial EEG, seizure detection, and adjustable stimulation parameters. The hardware, embedded on a small printed circuit board, is tested in C57BL/6 mice with drug-induced seizures and optogenetic stimulation, demonstrating its ability to suppress seizures effectively.

- [21]. Laura M. Ferrari, Usein Ismailov, Jean-Michel Badier, Francesco Greco, Esma Ismailova, “*Conducting polymer tattoo electrodes in clinical electro- and magnetoencephalography*”, npj Flexible Electronics, 2020.

This study evaluates inkjet-printed conducting polymer tattoo electrodes for clinical EEG and MEG. These dry, flexible electrodes offer conformal, high-quality, long-term recordings. The research analyzes skin/electrode impedance and demonstrates the feasibility of high-density recordings for localizing neuropathological activities, showcasing their potential for advanced brain monitoring and diagnostics.

- [22]. Manuel Reis Carneiro, Aníbal T. de Almeida, Mahmoud Tavakoli, “*Wearable and Comfortable e-Textile Headband for Long-Term Acquisition of Forehead EEG Signals*”, *IEEE Sensors Journal*, Vol. 20, No. 24, December 15, 2020.

This paper introduces a textile-based EEG headband designed for comfort, ease of use, and integration with human-machine interfaces (HMI). The headband is made from a multi-layer stretchable e-textile that interfaces with printed electrodes on the skin and a rigid PCB island. The electrodes are screen-printed with conductive stretchable ink and offer low skin-electrode impedance, comparable to traditional Ag/AgCl electrodes. The system reduces setup complexity by eliminating individual electrode placement and wiring, and features integrated amplification and EMF noise rejection. The headband is suitable for applications in HMI control and sleep data acquisition, advancing the potential for daily-use EEG systems.

- [23]. Kun-Peng Gao, Gen-Cai Shen, Nan Zhao, Chun-Peng Jiang, Bin Yang, Jing-Quan Liu, “*Wearable Multifunction Sensor for the Detection of Forehead EEG Signal and Sweat Rate on Skin Simultaneously*”, *IEEE Sensors Journal*, Vol. 20, No. 18, September 15, 2020.

This paper presents a multifunctional wearable sensor for simultaneous EEG and sweat rate measurement. The sensor uses conductive fabric electrodes made of metallic silver (Ag) for low contact impedance and noise in EEG recording, with data captured through an ADS1299 converter. Sweat permeates through the fabric, and the sensor accelerates evaporation using microporous PDMS and CaCl₂ desiccant, creating a humidity gradient to estimate sweat rate. Relative humidity is measured using an embedded SHT20 chip, allowing continuous sweat rate tracking alongside EEG monitoring. The sensor wirelessly transmits both data types, offering a compact solution for dual physiological monitoring.

- [24]. Erwin Habibzadeh, Tonekabony Shad, Marta Molinas, Trond Ytterdal, “*Impedance and Noise of Passive and Active Dry EEG Electrodes: A Review*”, *IEEE Sensors Journal*, Vol. 20, No. 24, December 15, 2020.

This review examines the challenges and advancements in dry EEG electrodes, particularly focusing on impedance and noise issues compared to wet electrodes. While

dry electrodes offer benefits such as no skin preparation and suitability for long-term use, their impedance must align with EEG amplifiers to ensure accurate signal recording and minimize noise. The review discusses various electrode materials and amplifier designs, comparing them to wet electrodes and industry standards. It identifies key challenges in dry electrode development and offers guidelines for future improvements, highlighting the need for further advancements to meet clinical requirements.

[25]. Yu M. Chi, Patrick Ng, Kang Eric, Joseph Kang, Jennifer Fang, “Wireless Non-contact Cardiac and Neural Monitoring”, 2010.

This research addresses the limitations of traditional wet adhesive electrodes for wireless ECG/EEG monitoring by introducing a non-contact capacitive electrode system. The system, which eliminates the discomfort of skin contact, is embedded within fabric layers for comfortable, unobtrusive use. The study provides detailed discussions on the design of high-performance capacitive sensors, including circuit schematics and construction techniques. The non-contact electrodes are integrated into wearable ECG chest harnesses and EEG headbands, paired with a compact, battery-powered wireless data acquisition system for real-time monitoring of cardiac and neural signals. Experimental results demonstrate that these capacitive electrodes perform similarly to conventional Ag/AgCl electrodes, with the wearable harnesses ensuring stable, movement-free electrode positioning. This development could play a significant role in advancing wireless, long-term physiological monitoring.

CHAPTER 4

WIRELESS ELECTRODE FOR BIOSIGNAL ACQUISITION

4.1. SYSTEM MODELLING

This project focuses on developing a portable, wireless EEG acquisition system that delivers precise, real-time monitoring of brain activity in various settings. The system replaces conventional wired EEG configurations with a wireless electrode approach (Figure 3), enabling high-quality signal capture without the limitations imposed by cables. At the core of the design, electrodes positioned on the scalp detect microvolt-level neural signals, which are then interfaced with an analog front-end (AFE) circuit for signal conditioning.

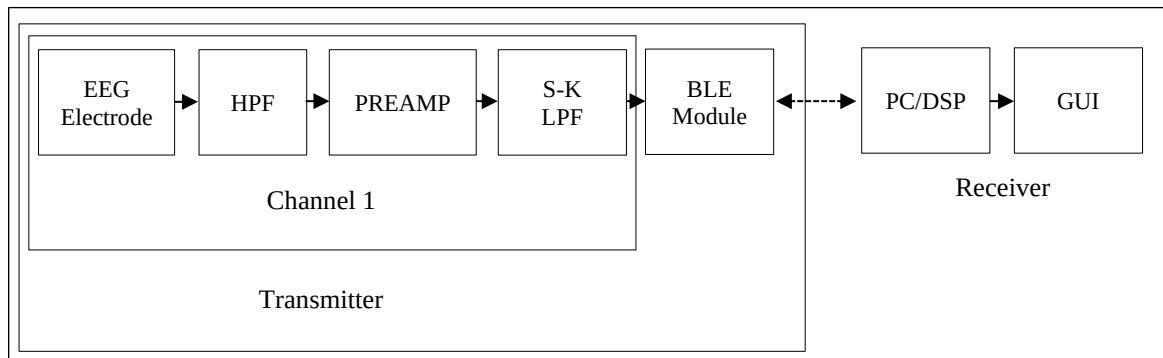


Figure 3. The block diagram of the system.

Within the AFE, the initial stage consists of an RC high-pass filter that employs a 330 nF capacitor in series with the electrode. When combined with the inherent $1\text{ M}\Omega$ electrode–skin impedance, this filter effectively blocks DC components and low-frequency noise, ensuring that only the relevant EEG signals pass through. The filtered signal is then amplified by a preamplifier stage, boosting the weak EEG signals to levels suitable for further processing. Next, a Sallen–Key low-pass filter—constructed with precision resistors and capacitors—further refines the signal by attenuating high-frequency noise and limiting the bandwidth to the EEG frequency range, thereby enhancing both the signal-to-noise ratio (SNR) and the common-mode rejection ratio (CMRR).

Once conditioned, the analog signals are routed to an analog-to-digital converter (ADC) integrated within a Bluetooth Low Energy (BLE) module. The ADC digitizes these

signals, and the resulting digital data is wirelessly transmitted to a host device, eliminating the need for cumbersome wiring and allowing for flexible, real-time monitoring. On the receiving end, a computer or host device equipped with a Bluetooth receiver captures the transmitted data, which is then processed and visualized using a Python-based application. This application provides a user-friendly graphical interface for the real-time monitoring and analysis of brain activity.

Overall, this project demonstrates an integrated approach to wireless EEG acquisition by combining advanced signal conditioning with efficient digital processing techniques. The resulting scalable, low-power solution enhances patient mobility and comfort while ensuring high-fidelity data capture, thereby supporting a range of applications including clinical diagnostics, neurofeedback, brain-computer interfaces (BCIs), and cognitive state monitoring.

4.1.1. CIRCUIT DIAGRAM

In the initial design phase, an AD623 instrumentation amplifier was employed as the primary amplification stage in a purely wired configuration to facilitate the study of various design parameters. Subsequently, a high-pass filter (HPF) with a cutoff frequency of 0.5 Hz was introduced, followed by further amplification using an AD8603 operational amplifier and a low-pass filter (LPF) with a cutoff frequency of approximately 60 Hz . The output from the LPF was directly monitored using a digital storage oscilloscope (DSO), allowing for detailed evaluation of the signal conditioning performance.

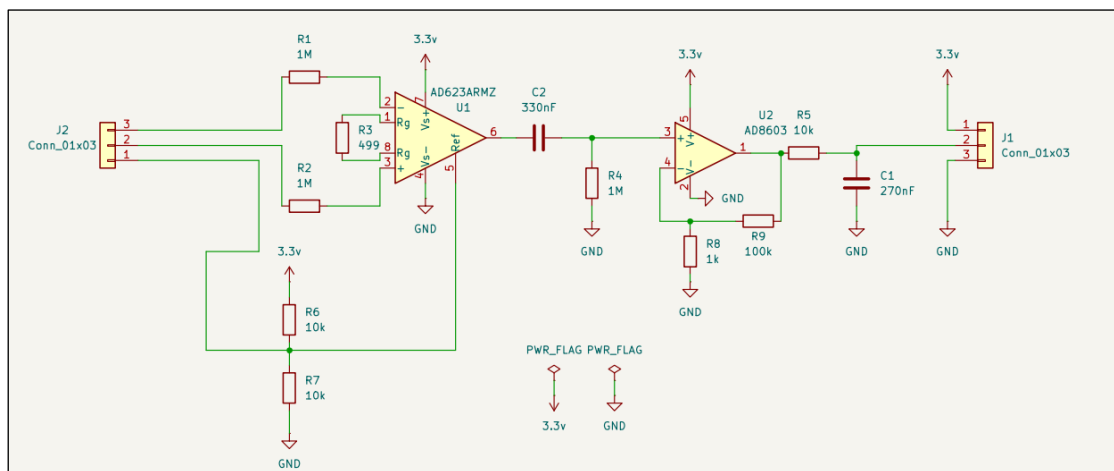


Figure 4. Initial design schematic of the EEG acquisition circuit.

The final circuit design begins at the electrode array, where multiple electrodes capture low-amplitude EEG signals from the scalp. Each electrode is AC coupled via a 330 nF capacitor, and in combination with the approximately $1\text{ M}\Omega$ electrode–skin impedance, this forms an RC high-pass filter that blocks DC offsets and slow baseline drift. The cutoff frequency of this filter is given by:

$$f_c = \frac{1}{2\pi RC} \approx \frac{1}{2\pi \times 1\text{M}\Omega \times 330\text{nF}} \approx 0.48\text{ Hz}$$

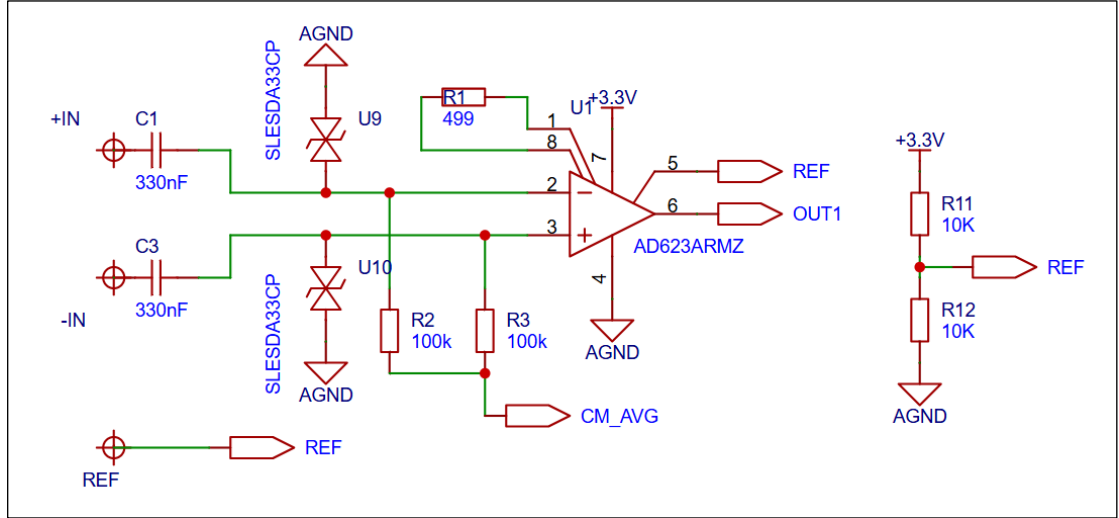


Figure 5. The input stage with high-pass filter, preamp and circuit protection

At each electrode input, low-capacitance TVS diodes (such as the SLESDA33CP) are implemented to provide robust ESD protection. These diodes clamp voltage spikes, ensuring that transient voltages do not damage sensitive circuit components, thereby preserving signal integrity. Following the filtering and ESD protection stages, the signals are fed into an AD623 instrumentation amplifier. A precision $500\ \Omega$ resistor is connected between the gain-setting pins of the AD623 to achieve a gain calculated by:

$$Gain = 1 + \frac{50\text{k}\Omega}{R_G} \approx \frac{50\text{k}\Omega}{500\Omega} \approx 101$$

resistor divider composed of two $10\text{ k}\Omega$ resistors provides a stable 1.65 V bias to the AD623's reference (REF) pin, centering the output at mid-supply (1.65 V) and ensuring high common-mode rejection. The amplified signal is then routed to an active filtering stage built around the AD8603 operational amplifier. This stage is split into two sections. The first section is a Sallen–Key low-pass filter, designed to remove high-frequency noise and confine the signal to the EEG frequency range.

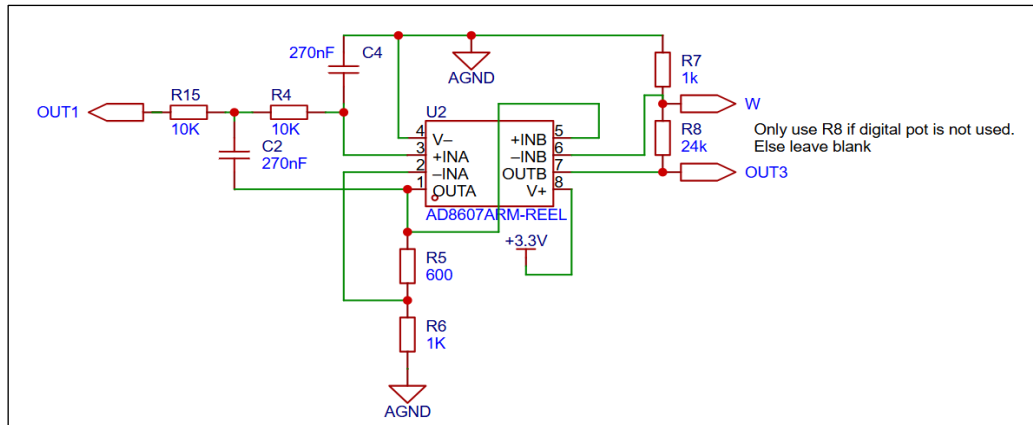


Figure 6. The second stage amplifier with sallan-key low pass filter, third stage amplifier

Two 10 k Ω resistors and two 270 nF capacitors are used to set the cutoff frequency:

$$f_c = \frac{1}{2\pi \times 10k\Omega \times 270nF} \approx 59 \text{ Hz}$$

This low-pass filter not only cleans the signal but also enhances the overall signal-to-noise ratio (SNR) and common-mode rejection ratio (CMRR). The second section is a non-inverting amplifier stage, where a fixed 1 k Ω resistor is paired with a digitally adjustable resistor implemented via an AD5165 digital potentiometer. For example, with a full-scale resistance near 24 k Ω , the gain of this stage is:

$$\text{Gain} = 1 + \frac{24\Omega}{1k\Omega} \approx 25$$

Combined with the Sallen–Key filter’s gain (approximately 1.6), the overall gain of the active stage is about 40. When cascaded with the AD623’s gain of ~ 101 , the complete signal chain yields an overall amplification of roughly 4040, making microvolt-level EEG signals readily measurable. To further mitigate common-mode interference, a Driven Right Leg (DRL) circuit is incorporated. High-value resistors (typically 100 k Ω) average the signals from the active electrodes to derive a common-mode

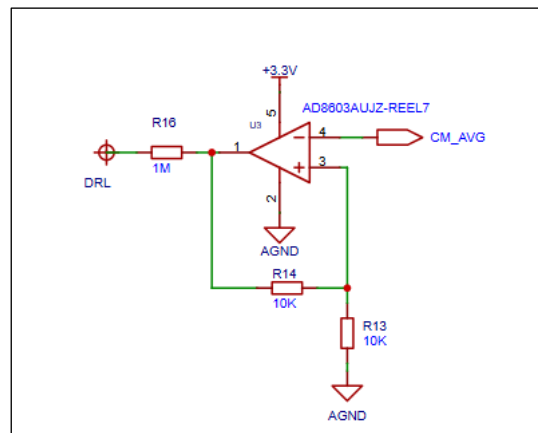


Figure 7. The DRL circuit implemented

Table 1. Components Selected.

No	Component	Specification
1	AD623	Instrumentation Amplifier, Low-power amplifier. Supply range ~2.7V to 12V, Gain adjustable.
2	AD8607	Low-power Operational Amplifier. Optimized for low noise and low distortion applications.
3	AD8603	Supply Range: 1.8V to 5V. Low noise performance (approximately $22\text{ nV}/\sqrt{\text{Hz}}$) suitable for high-fidelity signal conditioning.
4	AD5165	Digital Potentiometer, 50k, 255 stages. SPI communication.
5	Seeed Xiao nRF52840 BLE Module	Wireless Microcontroller, BLE transceiver. 32-bit MCU, 10-bit ADC.
6	Li-ion Battery	3.3V, 400mAh.
7	SLESDA33CP	Low-capacitance transient voltage suppression diode for robust ESD protection.

as the first stage of signal conditioning, boosting microvolt-level neural signals to levels that can be effectively processed by subsequent amplification stages.

Optimized for low noise and minimal distortion, the AD8607 is a low-power operational amplifier that plays a critical role in preserving signal integrity. Its performance ensures that any additional amplification does not introduce significant noise or distortion into the signal chain. This makes the AD8607 particularly useful in applications requiring precise signal processing, such as further conditioning of the EEG signal after the initial amplification provided by the AD623.

The AD8603 operates within a supply range of 1.8V to 5V and is renowned for its exceptionally low noise performance around $22\text{ nV}/\sqrt{\text{Hz}}$. This makes it highly suitable for high-fidelity signal conditioning in sensitive applications. Within the EEG circuit,

the AD8603 ensures that even after significant amplification, the EEG signals remain clean and free from extraneous noise, thus maintaining a high signal-to-noise ratio essential for accurate data acquisition.

The AD5165 is a digital potentiometer with a resistance of $50k$ and 255 tap positions, controlled via SPI communication. It allows for precise, digitally adjustable resistance, which is crucial for fine-tuning the gain settings within the signal conditioning circuit. By enabling dynamic adjustments of the amplification parameters, the AD5165 ensures that the system can adapt to varying signal conditions, thereby optimizing the performance and accuracy of the EEG measurements.

The Seeed Xiao nRF52840 BLE Module is a compact wireless microcontroller that integrates a Bluetooth Low Energy transceiver along with a 32-bit MCU and a built-in 10-bit ADC. This module is responsible for digitizing the conditioned analog EEG signals and transmitting the data wirelessly to a host device. Its low power consumption and compact form factor make it ideal for portable, battery-operated applications, ensuring real-time, wireless monitoring of brain activity without the constraints of wired connectivity.

A Li-ion battery rated at 3.3V and 400mAh is used to power the system. Its specifications are well-suited to the circuit's low-power components, providing a stable and reliable energy source that enables extended operation in portable applications. The battery's high energy density and compact size contribute significantly to the overall mobility and usability of the EEG acquisition system.

The SLESDA33CP is a low-capacitance transient voltage suppression diode designed for robust ESD protection. It protects sensitive circuit elements from voltage spikes and transient surges without compromising the high-speed, low-noise performance of the analog front-end. Its inclusion in the design ensures long-term reliability and preserves the integrity of the EEG signals by mitigating potentially damaging electrical disturbances.

In this design, resistors and capacitors form an integral part of the signal conditioning and filtering network, ensuring both stability and precision in the circuit. Resistors with values of $1M\Omega$, $100k\Omega$, $10k\Omega$, $1k\Omega$, 600Ω , and 499Ω are strategically employed for biasing, gain setting, and impedance matching. For instance, the $1M\Omega$ resistor, when paired with a 330 nF capacitor, creates a high-pass filter at the electrode-skin interface to block DC offsets and low-frequency noise, thereby isolating the relevant neural

signals. Other resistor values are utilized in voltage dividers and feedback loops to fine-tune amplifier gains and maintain proper operating conditions across different stages. Meanwhile, capacitors such as the 330 nF and 270 nF are crucial for AC coupling and filtering applications; the 330 nF capacitor aids in filtering out unwanted DC components, while the 270 nF capacitor helps in attenuating high-frequency noise. Together, these passive components work synergistically to enhance signal integrity and ensure that the microvolt-level EEG signals are accurately conditioned for subsequent processing and digitization.

4.3. DESCRIPTION OF DESIGNING SOFTWARE

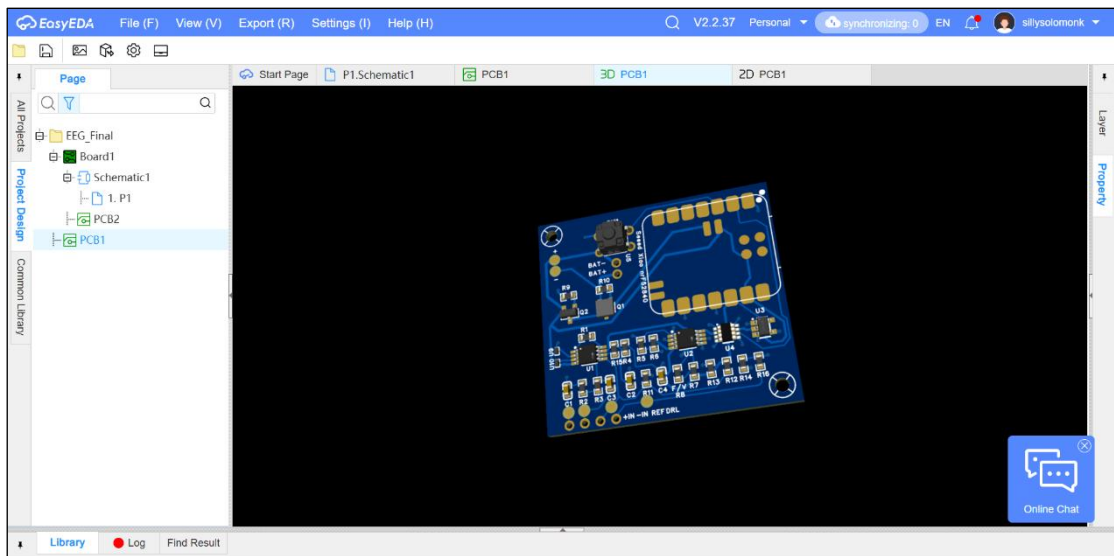


Figure 9. EasyEDA PCB designing window.

The circuit for the EEG transmitter system was designed using EasyEDA, a user-friendly, web-based electronic design automation (EDA) platform that streamlines schematic capture, simulation, and PCB layout. EasyEDA offers a comprehensive suite of tools that enabled detailed design and rigorous testing of the system's critical components, such as the AD623 instrumentation amplifier, AD8603 operational amplifier, Sseed Xiao nRF52840 BLE module, and other essential circuitry. Utilizing EasyEDA's simulation features, the design was thoroughly verified for electrical functionality, optimized component placement, and efficient routing, ensuring the circuit met low power consumption and performance requirements. This integrated approach allowed for accurate validation of the circuit's behavior before moving to physical prototyping, making EasyEDA an indispensable tool in the development of the project.

4.4. DESCRIPTION OF DATABASE

The EEG data used to train the brain state detection model and develop the Python-based programs and GUI for this project was sourced from PhysioNet, a widely recognized repository of physiological signals and health data. PhysioNet offers a diverse range of EEG datasets that are essential for developing and testing signal processing algorithms as well as creating user interfaces for monitoring and analyzing brain activity. By leveraging this extensive database, the project utilizes real-world EEG recordings to refine signal processing techniques, ensure system reliability, and optimize the GUI for real-time monitoring and analysis of brain signals. In addition, data collection from multiple pupils is being performed using the developed device for future studies, thereby providing a robust foundation for further research and validation of the system's performance.

4.5. DESCRIPTION OF PROTOTYPE DEVELOPMENT

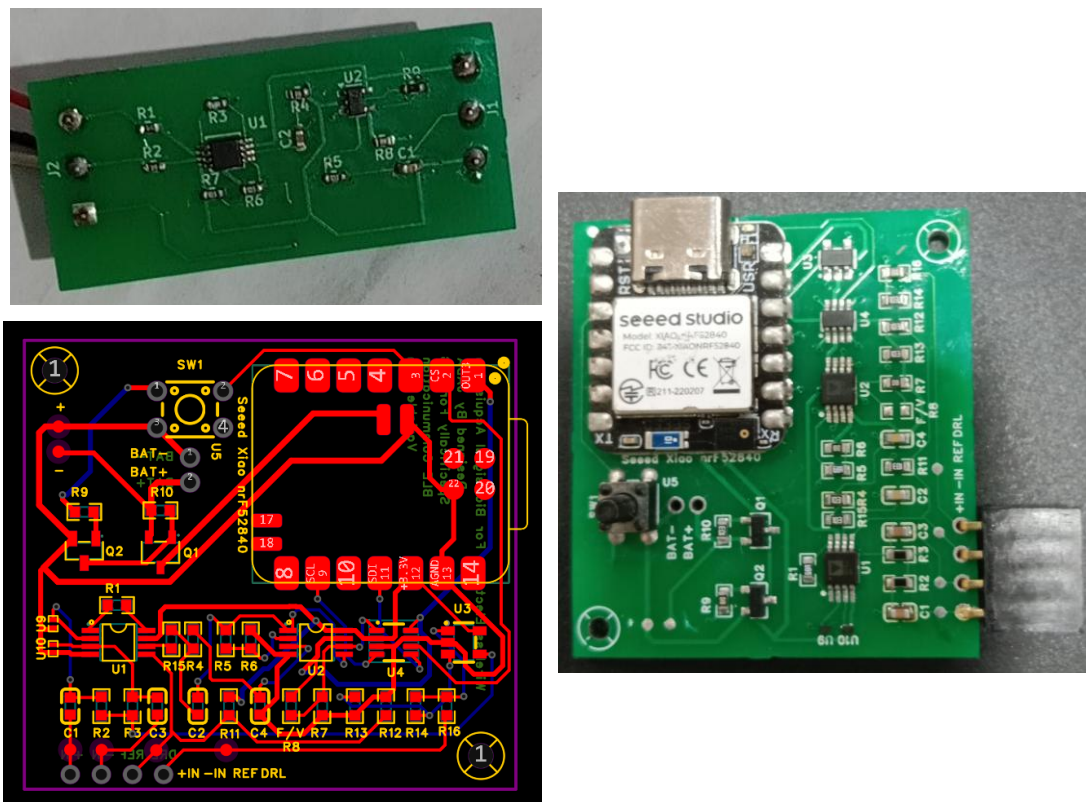


Figure 10. Initial EEG acquisition system without BLE, The Final PCB with BLE and layout

In the early stages of development, a non-wireless EEG acquisition system was implemented as a proof of concept. This initial prototype utilized the AD623 instrumentation amplifier and the AD8603 operational amplifier to capture and

condition EEG signals. At this stage, the design did not incorporate a Sallen–Key low-pass filter, and the gain was fixed rather than adaptively controlled. Although the early prototype successfully demonstrated basic signal acquisition and conditioning, these limitations prompted further refinement. In the final PCB design, these issues were addressed by integrating a Sallen–Key low-pass filter for enhanced noise reduction and incorporating a digitally adjustable gain stage using a digital potentiometer. Additionally, wireless communication was enabled through a Bluetooth Low Energy (BLE) module, thereby overcoming the constraints of the wired setup and significantly enhancing the system’s portability and overall functionality.

The final PCB is a highly integrated design that offers enhanced performance and portability for EEG acquisition. It features a Sallen–Key low-pass filter to effectively attenuate high-frequency noise and a digitally adjustable gain stage via the AD5165 for adaptive amplification of microvolt-level EEG signals. Wireless communication is enabled by the Seeed Xiao nRF52840 BLE module, allowing for real-time data transmission without the need for wired connectivity. The system is powered by an integrated Li-ion battery, which ensures extended operational time, and is protected by robust ESD protection through the SLESDA33CP TVS diode. Additionally, a driven right leg (DRL) circuit is incorporated to reduce common-mode interference, further enhancing signal quality.

Complementing the advanced analog front-end, the PCB includes a multifunction button that provides convenient control over the MCU. Real-time feedback of the connection state is delivered via an RGB LED, while the MCU firmware is designed to handle errors effectively, ensuring reliable operation throughout. This comprehensive design results in a low-power, high-performance solution optimized for EEG data acquisition and monitoring.

4.6. DESCRIPTION OF THE EEG DESKTOP APPLICATION

The EEG desktop application is a Python-based tool designed to provide a comprehensive and intuitive interface for real-time monitoring and analysis of brain activity. It seamlessly collects EEG signals from connected sensors via BLE connectivity, ensuring continuous data acquisition without the need for cables. The application offers dynamic visualization tools, including real-time waveforms and spectrograms, enabling users to closely examine and interpret the captured signals

along with the brain state display field, which works based on the trained model based on the dataset from PhysioNet. Additionally, users can adjust the sampling rate to balance data resolution with resource usage and control the digital potentiometer to fine-tune signal characteristics directly through the interface.

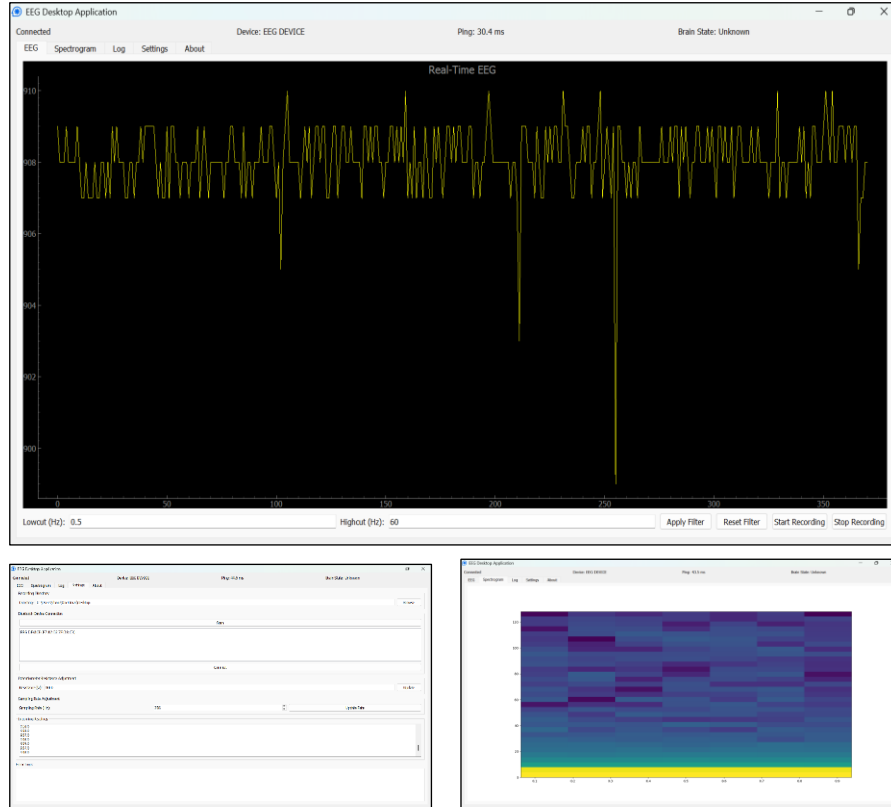


Figure 11. Different tabs in EEG Desktop Application a python-based GUI

Beyond real-time monitoring, the application features robust data management capabilities. Users can annotate events during EEG recording sessions for detailed analysis, and all EEG data is logged and can be exported in formats compatible with further review. The inclusion of user authentication mechanisms ensures data privacy and security, while the firmware update feature allows users to keep the connected EEG hardware current with the latest functionalities and improvements. Together, these features create a powerful and user-friendly EEG monitoring tool that significantly enhances data analysis and overall user experience.

4.7. DESCRIPTION ON THE VALIDATION SETUP

The validation setup for the EEG acquisition system utilized the BioAmp EXG module as a reference to assess and benchmark our device's performance. Although the BioAmp EXG module does not offer clinical-grade accuracy, it provided a practical

baseline for comparison. In this setup, both our device and the BioAmp EXG module were connected to a standard electrode array on a test subject, allowing for the

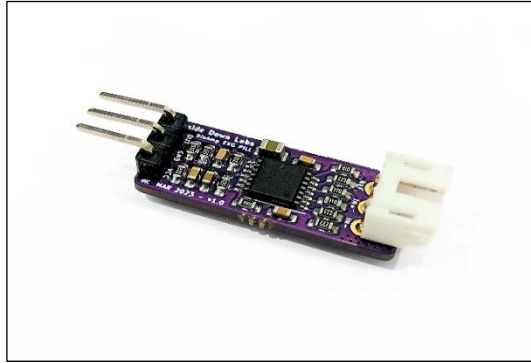


Figure 12. BioAmp EXG Pill

simultaneous capture of EEG signals. Key performance metrics including signal amplitude, noise levels, and common-mode rejection were evaluated by comparing the data from both systems.

The results of this comparative analysis revealed that our device not only met the expected performance standards but, in several aspects, demonstrated superior signal conditioning and transmission capabilities. The advanced filtering and adaptive gain features of our design contributed to a clearer, higher-fidelity EEG signal, even when benchmarked against the BioAmp EXG module. Additionally, the reliable wireless data transmission via the BLE module further confirmed the robustness of our system. Overall, despite the non-clinical grade accuracy of the BioAmp EXG module, it served as an effective reference, and the validation process underscored our device's enhanced performance in capturing and transmitting EEG data.

CHAPTER 5

DISCUSSIONS

The progress achieved in the development of the EEG transmitter circuit underscores the effectiveness of a design that emphasizes low power consumption and portability key requirements for wearable, continuous EEG monitoring. The system is engineered to capture, condition, and wirelessly transmit EEG signals with high fidelity while operating on a rechargeable battery for extended durations. The current phase of the project involves finalizing the PCB design and fabrication, followed by meticulous component assembly and rigorous testing to ensure optimal signal acquisition, processing, and wireless transmission. Concurrently, the development of advanced signal processing models in a Python-based environment and a dedicated GUI for real-time control and monitoring further refines the system's performance and usability.

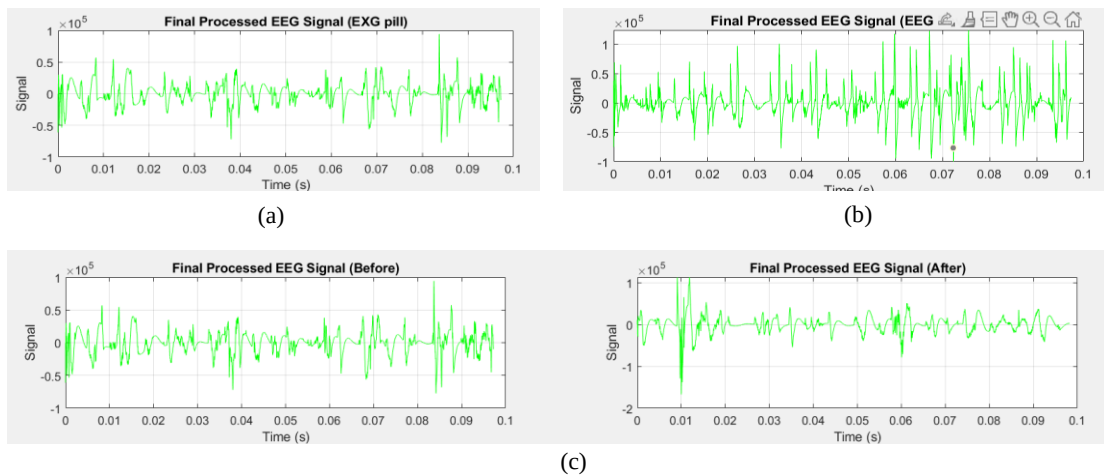


Figure 13.(a), (b) EEG reading using BioAmp EXG pill and EEG Device, (c) EEG reading Before and after the use of CES machine.

Validation of the system was conducted through a comparative analysis with the BioAmp EXG Pill, which, although not providing clinical-grade accuracy, served as an effective benchmark. Our device consistently demonstrated a significantly higher signal-to-noise ratio (SNR) and superior overall performance, while having a comparable Common Mode. The Common Mode is analyzed by comparing the signal reading from both the EXG Pill and EEG Device. The SNR was calculated by using RMS value of the raw signal with the filtered signal. This comparison showed a significant superiority in the SNR of EEG Device. Additionally, the correlation factor between the BioAmp EXG Pill and our EEG device was measured, yielding an average

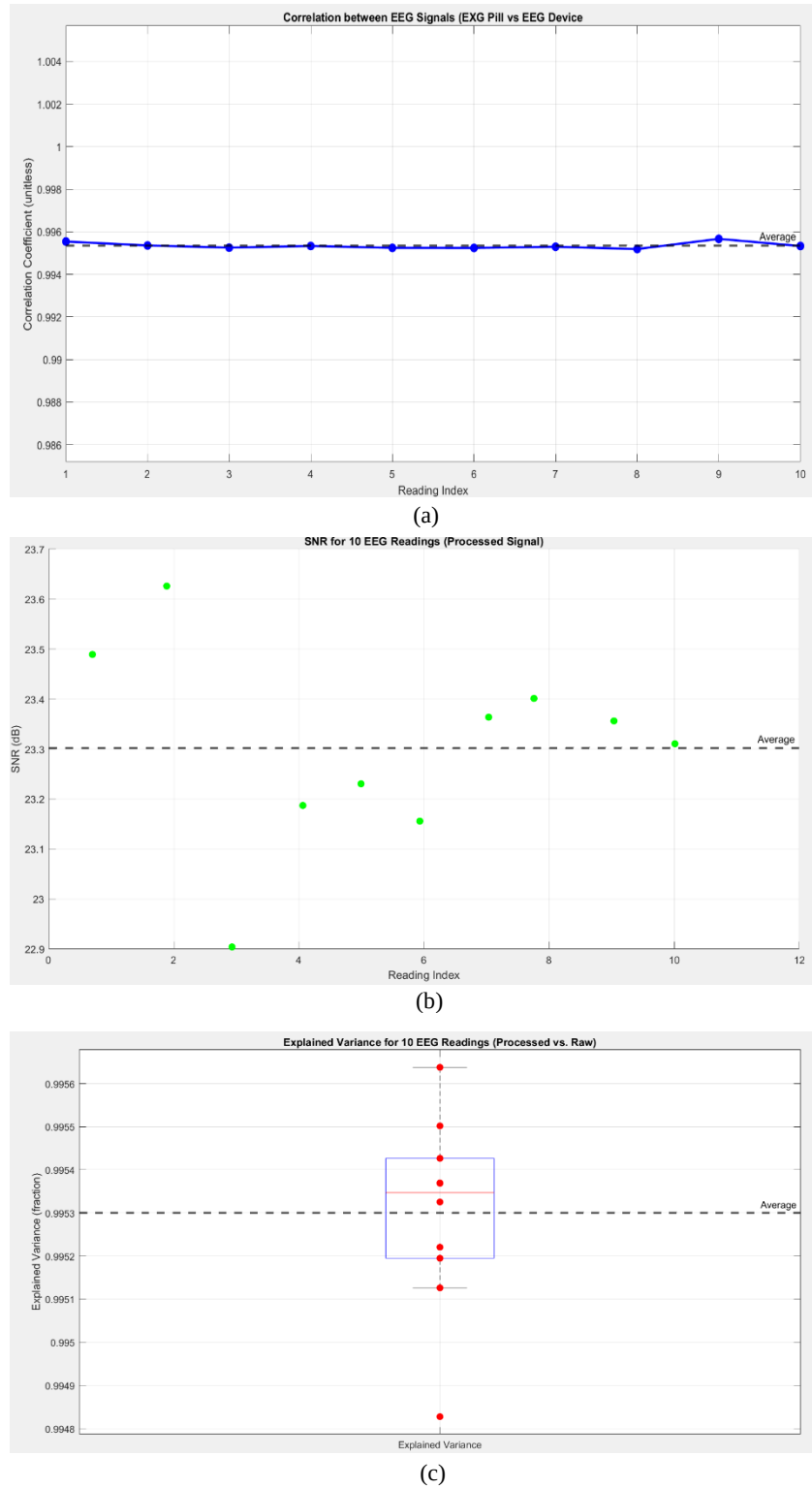


Figure 14. (a) Correlation Factor between the EXG Pill and Our EEG Device for 10 different readings, (b) SNR for 10 different readings, (c) Explained Variance of the EEG readings

value exceeding 0.99, indicating a high degree of agreement between the two devices (Figure.14 a). Furthermore, the signal-to-noise ratio (SNR) of our device's signal was found to be greater than 20 dB in most cases, reflecting the system's capability to capture high-quality signals with minimal noise (figure14. b). The explained variance

was nearly 1, suggesting that the device accounts for almost all the variability in the recorded EEG signals (Figure 14. c).

Recordings taken from subjects before and after the application of Cranial Electrical Stimulation (CES) revealed that post-CES signals were notably calmer with reduced peak amplitudes, indicating the device's sensitivity to changes in neural activity. These findings not only confirm that our design outperforms the reference module in terms of SNR and signal clarity but also validate its potential as a robust platform for advanced EEG monitoring applications. Validation of the system was conducted through a comparative analysis with the BioAmp EXG Pill, which, although not providing clinical-grade accuracy, served as an effective benchmark. Our device consistently demonstrated a significantly higher signal-to-noise ratio (SNR) and superior overall performance, while having a comparable Common Mode. The Common Mode is analyzed by comparing the signal reading from both the EXG Pill and EEG Device. The SNR was calculated by using RMS value of the raw signal with the filtered signal. This comparison showed a significant superiority in the SNR of EEG Device. Additionally, the correlation factor between the BioAmp EXG Pill and our EEG device was measured, yielding an average value exceeding 0.99, indicating a high degree of agreement between the two devices. Furthermore, the signal-to-noise ratio (SNR) of our device's signal was found to be greater than 20 dB in most cases, reflecting the system's capability to capture high-quality signals with minimal noise. The explained variance was nearly 1, suggesting that the device accounts for almost all the variability in the recorded EEG signals. Recordings taken from subjects before and after the application of Cranial Electrical Stimulation (CES) revealed that post-CES signals were notably calmer with reduced peak amplitudes, indicating the device's sensitivity to changes in neural activity (figure 13. c). These findings not only confirm that our design outperforms the reference module in terms of SNR and signal clarity but also validate its potential as a robust platform for advanced EEG monitoring applications.

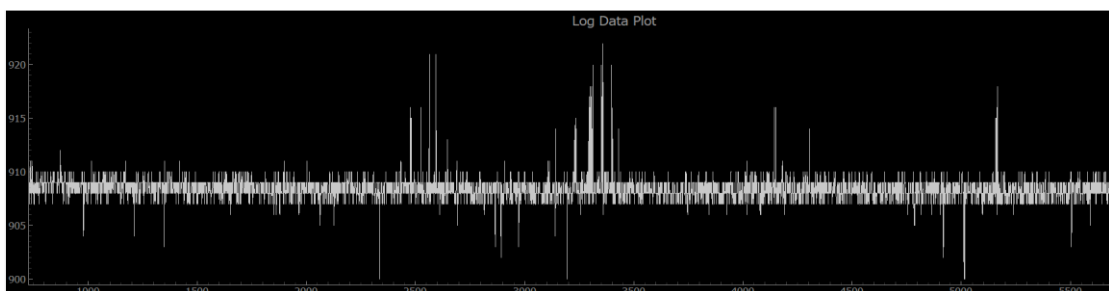


Figure 15. EEG reading for resting and cognitive state using EEG Device.

The effectiveness of the developed EEG transmitter system was further evaluated through real-time monitoring of brain activity in various cognitive states (Figure 15). The study measured EEG signals during rest, active engagement, and a return to the rest state, demonstrating the device's ability to distinguish between these conditions. Notably, peak EEG signals were observed during the active state, particularly when subjects were engaged in gaming activities. This aligns with expectations, as heightened cognitive engagement and mental workload typically result in increased neural activity. A clear distinction in EEG patterns between the resting and active states was evident, with significantly higher amplitude fluctuations during periods of intense engagement. This real-time analysis capability, enabled by the wireless EEG acquisition system, reinforces the utility of the device for applications requiring continuous monitoring of cognitive states. The system's ability to provide reliable data on fluctuations in brain activity showcases its robustness for applications in neuroscience research, neurofeedback training, and potential cognitive state monitoring.

Beyond validating the system's performance, the results demonstrate its effectiveness in detecting cognitive state changes, confirming its suitability for real-world applications. The combination of high SNR, adaptive signal acquisition, and wireless transmission ensures that this system can be a valuable tool in studying dynamic brain activity patterns, distinguishing it from conventional wired EEG setups. By integrating these findings with further refinements in signal processing and hardware optimization, the developed system holds promise for broader use in various domains, including mental health monitoring, cognitive load assessment, and human-computer interaction research.

CHAPTER 6

CONCLUSION

From a technology development standpoint, our project validates the concept of a compact, low-power, portable, and wireless adaptive EEG acquisition system designed to capture and process microvolt-level brain signals with exceptional fidelity. The system's architecture, which integrates an optimized analog front-end with advanced digital gain control and robust ESD protection, ensures reliable performance under dynamic conditions. The incorporation of a BLE module for real-time wireless data transmission allows the device to operate efficiently on battery power, making it well-suited for continuous, on-the-go monitoring. Rigorous comparative testing against conventional reference modules has shown that our device delivers a superior signal-to-noise ratio and overall signal clarity, confirming the robustness and functional reliability of our design.

Looking ahead, this project lays a strong foundation for future innovations in wearable neurotechnology. The system's adaptive and energy-efficient design not only meets current objectives but also provides a versatile platform for further development. Planned enhancements include refined firmware for improved error handling and adaptive gain tuning, as well as potential integration of neuromodulation features to broaden the device's application scope. Ultimately, these advancements position our system as a transformative solution for advanced EEG monitoring, neurofeedback, and brain-computer interface applications, setting the stage for continued progress in non-invasive, real-time brain signal analysis.

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APPENDIX

CODE USED:

1. FIRMWARE CODE:

```
#include <ArduinoBLE.h>
#include <SPI.h>

// ----- BLE Service and Characteristics -----
BLEService uartService("6E400001-B5A3-F393-E0A9-E50E24DCCA9E");
BLECharacteristic txCharacteristic("6E400003-B5A3-F393-E0A9-E50E24DCCA9E",
                                   BLERead | BLENotify, 20);
BLECharacteristic rxCharacteristic("6E400002-B5A3-F393-E0A9-E50E24DCCA9E",
                                   BLEWrite | BLEWriteWithoutResponse, 20);

// ----- Constants & Variables -----
#define ADC_PIN A0
#define POT_CS A1
#define BUTTON_PIN A2 // Tactile button on A2 (e.g., connected to ground)
const uint8_t MAX_POT_VALUE = 255;
const uint16_t MAX_RESISTANCE = 50000; // 50kΩ pot
const uint16_t DEFAULT_RESISTANCE = 24000; // 24kΩ default

uint32_t adcInterval = 100; // ADC sampling interval in ms (default)
uint16_t samplingRate = 256; // Default sampling rate in Hz

// ----- SPI Settings -----
SPISettings potSPISettings(1000000, MSBFIRST, SPI_MODE0);

// ----- LED State Management -----
enum LEDState {
    SYSTEM_ON,
    BLE_ADVERTISING,
    BLE_CONNECTED
};
```

```

void setLEDState(LEDState state) {
    // Assumes that LEDR, LEDB, LEDG are defined and configured as outputs
    switch(state) {
        case SYSTEM_ON:
            digitalWrite(LEDRL, LOW); // Red ON
            digitalWrite(LEDDB, HIGH); // Blue OFF
            digitalWrite(LEDG, HIGH); // Green OFF
            break;
        case BLE_ADVERTISING:
            digitalWrite(LEDRL, HIGH); // Red OFF
            digitalWrite(LEDDB, LOW); // Blue ON
            digitalWrite(LEDG, HIGH); // Green OFF
            break;
        case BLE_CONNECTED:
            digitalWrite(LEDRL, HIGH); // Red OFF
            digitalWrite(LEDDB, HIGH); // Blue OFF
            digitalWrite(LEDG, LOW); // Green ON
            break;
    }
}

// ----- Potentiometer Control -----
uint8_t resistanceToPotCode(long resistance) {
    return constrain(map(resistance, 0, MAX_RESISTANCE, 0, MAX_POT_VALUE),
0, MAX_POT_VALUE);
}

void setPotResistance(long resistance) {
    resistance = constrain(resistance, 0, MAX_RESISTANCE);
    uint8_t value = resistanceToPotCode(resistance);

    digitalWrite(POT_CS, LOW);
    SPI.beginTransaction(potSPISettings);

```

```

    SPI.transfer(value);
    SPI.endTransaction();
    digitalWrite(POT_CS, HIGH);

    Serial.print("Pot set to: ");
    Serial.print(resistance);
    Serial.println("Ω");
}

// ----- BLE Setup -----
void setupBLE() {
    BLE.setDeviceName("EEG DEVICE");
    BLE.setLocalName("EEG DEVICE");
    BLE.setAdvertisedService(uartService);

    uartService.addCharacteristic(txCharacteristic);
    uartService.addCharacteristic(rxCharacteristic);
    BLE.addService(uartService);

    txCharacteristic.writeValue("EEG Ready");
    BLE.advertise();
    setLEDState(BLE_ADVERTISING);
}

void setup() {
    Serial.begin(115200);
    // No waiting for Serial in this version

    setLEDState(SYSTEM_ON);
    delay(1000);

    pinMode(POT_CS, OUTPUT);
    digitalWrite(POT_CS, HIGH);
    SPI.begin();

```

```

setPotResistance(DEFAULT_RESISTANCE);

// Set tactile button pin as input with internal pull-up
pinMode(BUTTON_PIN, INPUT_PULLUP);

if (!BLE.begin()) {
    Serial.println("BLE init failed!");
    while(1);
}
setupBLE();
}

void handleSerialInput() {
    if (Serial.available()) {
        String inputLine = Serial.readStringUntil('\n');
        inputLine.trim();
        if (inputLine.length() == 0) return;

        if (inputLine.startsWith("POT:")) {
            long resistance = inputLine.substring(4).toInt();
            setPotResistance(resistance);
            Serial.print("Potentiometer set to: ");
            Serial.print(resistance);
            Serial.println("Ω");
        } else if (inputLine.startsWith("TIME:")) {
            uint32_t newInterval = inputLine.substring(5).toInt();
            adcInterval = newInterval;
            Serial.print("ADC interval updated to: ");
            Serial.print(adcInterval);
            Serial.println(" ms");
        } else if (inputLine.startsWith("RATE:")) {
            uint16_t newRate = inputLine.substring(5).toInt();
            if (newRate > 0) {
                samplingRate = newRate;
            }
        }
    }
}

```

```

        adcInterval = 1000 / newRate;
        Serial.print("Sampling rate updated to: ");
        Serial.print(newRate);
        Serial.println(" Hz");
    }
} else if (inputLine.equalsIgnoreCase("pot")) {
    Serial.print("Enter resistance (0-");
    Serial.print(MAX_RESISTANCE);
    Serial.println("Ω:");
} else {
    Serial.print("Unknown command: ");
    Serial.println(inputLine);
}
}
}

void loop() {
    // Check for button press (restart switch) at all times
    if (digitalRead(BUTTON_PIN) == LOW) { // Button pressed (grounded)
        delay(50); // Debounce delay
        if (digitalRead(BUTTON_PIN) == LOW) {
            Serial.println("Restart button pressed. Restarting...");
            NVIC_SystemReset(); // Software reset
        }
    }
}

BLEDevice central = BLE.central();

if (central) {
    Serial.print("Connected: ");
    Serial.println(central.address());
    setLEDState(BLE_CONNECTED);

    uint32_t lastAdcSend = millis();

```



```

while (central.connected()) {
    // Check for restart button while connected
    if (digitalRead(BUTTON_PIN) == LOW) {
        delay(50);
        if (digitalRead(BUTTON_PIN) == LOW) {
            Serial.println("Restart button pressed during connection. Restarting...");
            NVIC_SystemReset();
        }
    }
}

if (millis() - lastAdcSend >= adcInterval) {
    lastAdcSend = millis();
    int adcValue = analogRead(ADC_PIN);
    char buffer[6];
    sprintf(buffer, "%d", adcValue);
    txCharacteristic.writeValue(buffer);
    Serial.print("ADC: ");
    Serial.println(adcValue);
}

if (rxCharacteristic.written()) {
    String command = String((char*)rxCharacteristic.value()).substring(0,
rxCharacteristic.valueLength());
    command.trim();
    if (command.startsWith("POT:")) {
        long resistance = command.substring(4).toInt();
        setPotResistance(resistance);
    }
    if (command.startsWith("TIME:")) {
        uint32_t newInterval = command.substring(5).toInt();
        adcInterval = newInterval;
        Serial.print("ADC interval updated to: ");
        Serial.print(adcInterval);
        Serial.println(" ms");
    }
}

```

```
    if (command.startsWith("RATE:")) {
        uint16_t newRate = command.substring(5).toInt();
        if (newRate > 0) {
            samplingRate = newRate;
            adcInterval = 1000 / newRate;
            Serial.print("Sampling rate updated to: ");
            Serial.print(newRate);
            Serial.println(" Hz");
        }
    }
}

handleSerialInput();
}

Serial.println("Disconnected");
setLEDState(BLE_ADVERTISING);
}

handleSerialInput();
}
```